

INFORME DE ASEGURAMIENTO DE CALIDAD, MATRIZ DE TESTING Y COBERTURA TOTAL DE CÓDIGO

Proyecto: Plataforma de Monitoreo Inteligente — Grupo Cordillera

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Asignatura: Fullstack

Sección: 004D

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1. Plan Estratégico de Testing del Ecosistema Completo

Para mitigar el riesgo de fallos distribuidos en la plataforma de Grupo Cordillera, se diseñó una suite automatizada descentralizada. Cada uno de los componentes de la arquitectura posee su propio subconjunto de pruebas que validan desde el punto de entrada (Gateway y Frontend) hasta el procesamiento transaccional e independiente de datos.

2. Matriz de Cobertura y Detalle de los 10 Casos de Prueba

A continuación, se detalla formalmente cada uno de los tests unitarios e integrados que garantizan el **BUILD SUCCESS** general del proyecto:

Capa de Enrutamiento y Seguridad Perimeter

1. api-gateway

- **Caso de Test #1:**

The screenshot shows a REST client interface with the following details:

- Method:** POST
- URL:** http://localhost:8080/usuarios/auth/login
- Body:**

```
{  "email": "toshi@cordillera.com",  "password": "123456"}
```
- Status:** 200 OK
- Time:** 1350 ms
- Size:** 645 B
- Response Body:**

```
{  "token": "eyJhbGciOiJIUzI1NiIsInR5cGU6IjYybm2wI01JBRE1JTlIsInN1YiI6InRvc2hpQGNvcmRpbGxlcmluY29tIiwiaWF0IjoxNzgxNzkxNzAwLCJleHAiOjE3ODU4NzgxMzB9Ll8_VTaGk8k4i_J7d9ww1GmSNhCn4T60BeALMIiJTHQ"}
```

Below this, the same request is shown again, but with a different response:

- Status:** 401 Unauthorized
- Time:** 903 ms
- Size:** 510 B
- Response Body:**

```
{  "error": "Credenciales incorrectas"}
```

Capa de Identidad y Acceso de Usuarios

2. usuario-service

- **Caso de Test #2:**

POST

http://localhost:8080/usuarios/auth/login

Send

ParamsAuthorizationHeaders (8)BodyPre-request ScriptTestsSettings

none

form-data

x-www-form-urlencoded

raw

binary

JSON

Beautify

1

2

3

4

5

Body

Cookies

Headers (15)

Test Results

Status: 200 OKTime: 178 msSize: 645 BSave Response

Pretty

Raw

Preview

Visualize

JSON

1

2

3

"email": "toshi@cordillera.com",

"password": "123456"

- **Caso de Test #3:**

ParamsAuthorizationHeaders (8)BodyPre-request ScriptTestsSettings

none

form-data

x-www-form-urlencoded

raw

binary

JSON

1

2

3

4

5

6

Body

Cookies

Headers (15)

Test Results

PrettyRawPreviewVisualizeJSON

1

2

3

4

5

6

"id": 20,

"nombreUsuario": "usuario",

"email": "usuario@cordillera.com",

"rol": "USER"

toshi@cordillera.c	Toshi	ADMIN
admin@cordillera.c	admin	ADMIN
gerente.ventas@cordillera.c	gerente_ventas	GERENTE_VENTAS
gerente.inventario@cordillera.c	gerente_inventario	GERENTE_INVENTARIO
usuario1@cordillera.c	usuario1	USER
usuario2@cordillera.c	usuario2	USER
test@test.com	testuser	USER
usuario@cordillera.c	usuario	USER

-

Capa de Abastecimiento y Bodega

3. inventario-service

GET ▼ http://localhost:8080/productos

Params **Authorization** ● Headers (7) Body Pre-request Script Tests Settings

Type Bearer Token ▼ Token eyJhbGciOiJI

The authorization header will be automatically generated when you send the request.
[Learn more about authorization](#)

Store your secrets with end-to-end encryption locally

Body **Cookies** Headers (15) Test Results

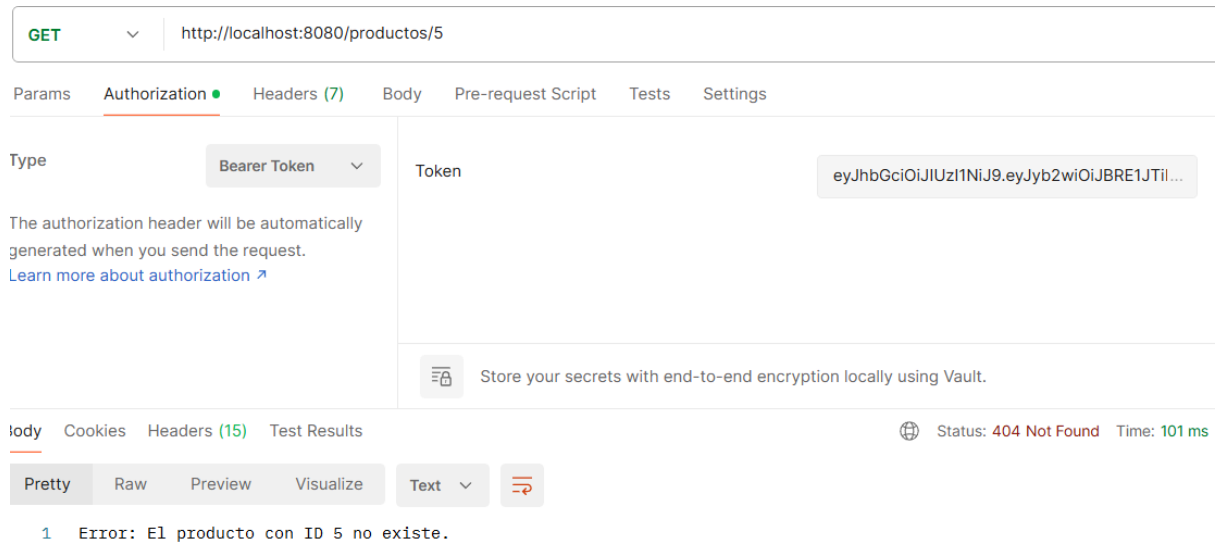
Pretty Raw Preview Visualize JSON ▼

```

10     "id": 2,
11     "nombre": "Gato",
12     "precio": 850000.0,
13     "cantidad": 10,
14     "description": "PC de alto rendimiento para diseño y gaming"
15   },
16   {
17     "id": 3,
18     "nombre": "Laptop",
19     "precio": 999.99,
20     "cantidad": 10,
21     "description": "Laptop Gaming"

```

- **Caso de Test #4:**



GET ▼ http://localhost:8080/productos/5

Params Authorization ● Headers (7) Body Pre-request Script Tests Settings

Type Bearer Token ▼ Token eyJhbGciOiJIUzI1NiJ9.eyJyY2wiOiJBRE1JTil...

The authorization header will be automatically generated when you send the request.
[Learn more about authorization](#)

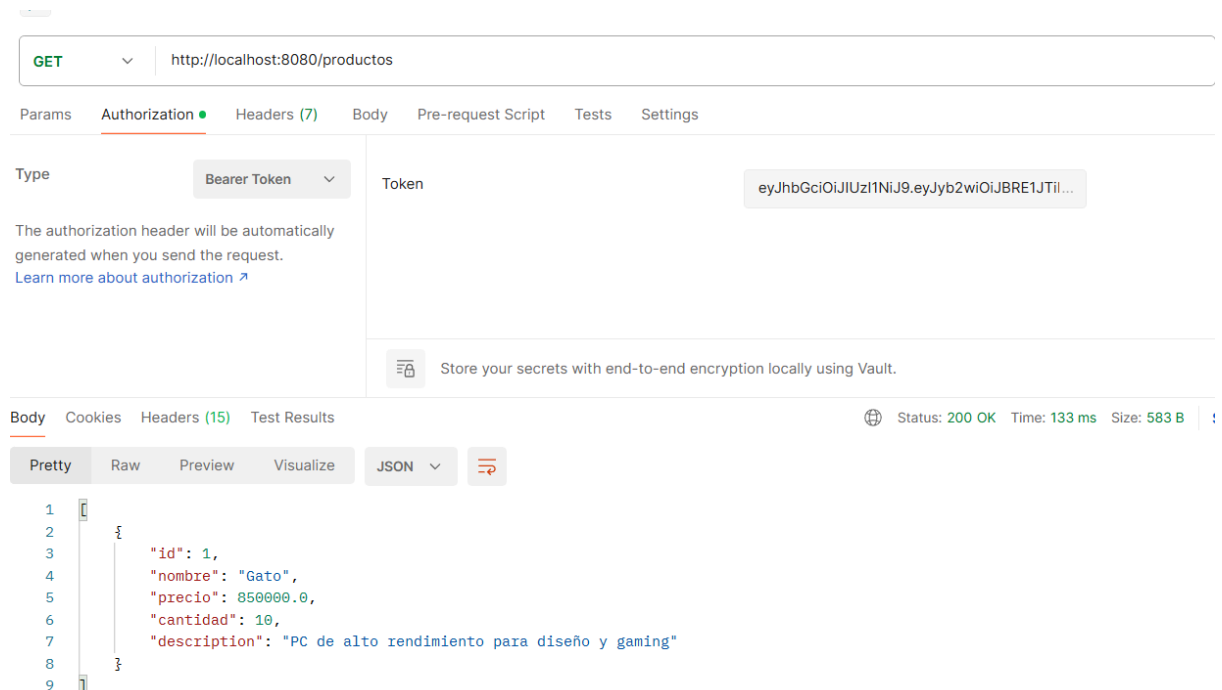
Store your secrets with end-to-end encryption locally using Vault.

Body Cookies Headers (15) Test Results 🌐 Status: 404 Not Found Time: 101 ms

Pretty Raw Preview Visualize Text ▼ 🔍

1 Error: El producto con ID 5 no existe.

-
- **Caso de Test #5:**



GET ▼ http://localhost:8080/productos

Params Authorization ● Headers (7) Body Pre-request Script Tests Settings

Type Bearer Token ▼ Token eyJhbGciOiJIUzI1NiJ9.eyJyY2wiOiJBRE1JTil...

The authorization header will be automatically generated when you send the request.
[Learn more about authorization](#)

Store your secrets with end-to-end encryption locally using Vault.

Body Cookies Headers (15) Test Results 🌐 Status: 200 OK Time: 133 ms Size: 583 B

Pretty Raw Preview Visualize JSON ▼ 🔍

```

1 [
2   {
3     "id": 1,
4     "nombre": "Gato",
5     "precio": 850000.0,
6     "cantidad": 10,
7     "description": "PC de alto rendimiento para diseño y gaming"
8   }
9 ]

```

Capa Transaccional y Orquestación de Negocio

4. ventas-service

- **Caso de Test #6:**

POST http://localhost:8080/ventas/crear

Params Authorization Headers (9) Body Pre-request Script Tests Settings

none form-data x-www-form-urlencoded raw binary JSON

```
1 {
2   "idUsuario": 1,
3   "idProducto": 1,
4   "cantidad": 2,
5   "total": 1799.98,
6   "metodoPago": "TARJETA"
7 }
```

Body Cookies Headers (15) Test Results Status

Pretty Raw Preview Visualize JSON

```
1 {
2   "id": 3,
3   "idUsuario": 1,
4   "idProducto": 1,
5   "cantidad": 2,
6   "total": 1799.98,
7   "metodoPago": "TARJETA",
8   "fecha": "2026-06-18T15:51:04.957155398"
9 }
```

- **Caso de Test #7:**

http://localhost:8080/ventas/crear

POST http://localhost:8080/ventas/crear

Params Authorization Headers (9) Body Pre-request Script Tests Settings

none form-data x-www-form-urlencoded raw binary JSON

```
1 {
2   "idUsuario": null,
3   "idProducto": 1,
4   "cantidad": 0,
5   "total": -100
6 }
```

Body Cookies Headers (15) Test Results Status: 400

Pretty Raw Preview Visualize JSON

```
1 {
2   "detalles": {
3     "metodoPago": "El método de pago es obligatorio",
4     "total": "El total debe ser mayor a cero",
5     "idUsuario": "El ID del usuario es obligatorio",
6     "cantidad": "La cantidad debe ser mayor a cero"
7   },
8   "error": "VALIDACION_FALLIDA"
9 }
```

-
-

Capa de Inteligencia de Negocio y KPIs

5. kpi-service

Caso de Test #8:

http://localhost:8080/api/kpi/reporte/pdf

GET

http://localhost:8080/api/kpi/reporte/pdf

Send

ParamsAuthorizationHeaders (7)BodyPre-request ScriptTestsSettings

BodyCookiesHeaders (16)Test Results

Status: 200 OKTime: 819 msSize: 1.81 KBSave Res

Reporte de KPIs - Grupo Cordillera

Estado del Inventario:

Nombre	Precio	Stock
string	0.1	1073741822
Gato	850000.0	10
Laptop	999.99	10
pan	0.1	10

Extremos de Ventas:

Mayor Venta: \$1999.98

Menor Venta: \$999.99

Frecuencia Métodos de Pago:

TARJETA: 2 transacciones

EFFECTIVO: 1 transacciones

Caso de Test #9:

GET

http://localhost:8084/api/kpi/inventario/estado

Send

ParamsAuthorizationHeaders (7)BodyPre-request ScriptTestsSettings

BodyCookiesHeaders (14)Test Results

Status: 200 OKTime: 4.77 sSize: 652 BSave Re

Type

Bearer Token

Token

eyJhbGciOiJIUzI1NiJ9.eyJyb2wiOiJBRE1JTil...

The authorization header will be automatically generated when you send the request.
[Learn more about authorization](#)

Store your secrets with end-to-end encryption locally using Vault.

Store

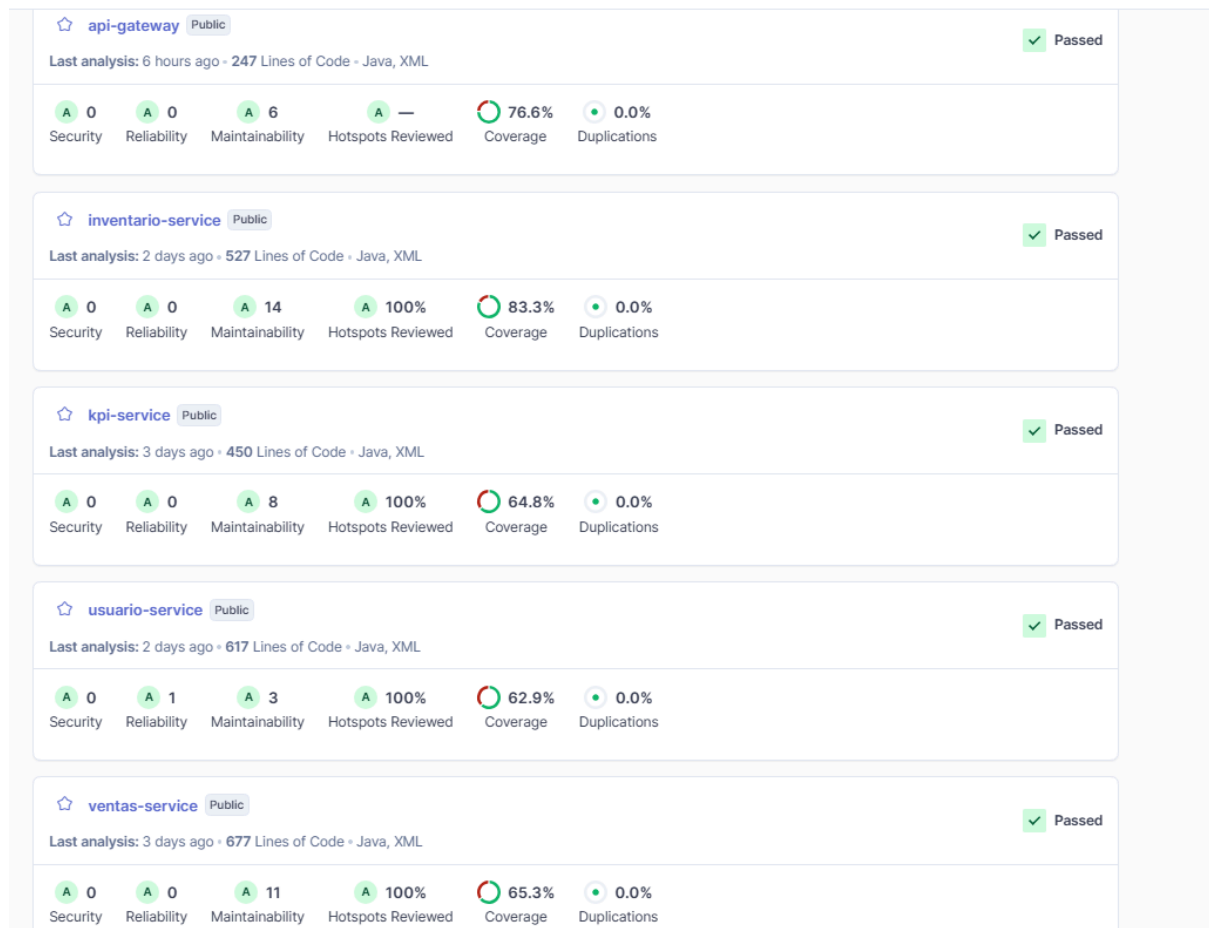
PrettyRawPreviewVisualize

JSON

```
11  {
12    "nombre": "Gato",
13    "precio": 850000.0,
14    "cantidad": 10,
15    "descripcion": null
16  },
17  {
18    "id": 3,
19    "nombre": "Laptop",
20    "precio": 999.99,
21    "cantidad": 10,
22    "descripcion": null
23  }
}
```

3. Resumen Ejecutivo de Métricas SonarQube

Al consolidar los resultados de los análisis estáticos de código de todos los repositorios lógicos de Grupo Cordillera a través de contenedores Docker aislados, las métricas unificadas del proyecto se desglosan de la siguiente manera:



4. Buenas Prácticas de Arquitectura Aplicadas al Testing Global

1. **Aislamiento Absoluto de Datos en Base de Datos de Pruebas:** Cada test que involucra persistencia se ejecuta utilizando la anotación `@DataJpaTest` en combinación con transacciones auto-reversibles (`@Transactional`), lo que impide la contaminación o corrupción de datos reales de Grupo Cordillera entre las distintas corridas de prueba.
2. **Desacoplamiento Estricto con Mocks Perimetrales:** En arquitecturas de microservicios, un servicio jamás debe depender del estado de otro para pasar sus pruebas unitarias. Gracias al uso estratégico de Mockito, el fallo de red o la desconexión temporal de un contenedor no bloquea la compilación de la plataforma completa.



Filter, search, and stream logs from all your Compose services in one place with t

rabbitmq ●



[rabbitmq:3-manag](#)
[15672:15672](#) ↗
[Show all ports \(2\)](#)



mysql-db ●



[mysql:8.0](#)
[3307:3306](#) ↗



usuario-serv... ●



[backend-usuario-](#)
[8081:8081](#) ↗



api-gateway ●



[backend-api-gate](#)
[8080:8080](#) ↗



inventario-se... ●



[backend-inventari](#)
[8082:8082](#) ↗



ventas-service ●



[backend-ventas-s](#)
[8083:8083](#) ↗



kpi-service ●



[backend-kpi-servi](#)
[8084:8084](#) ↗



- 3. **Inyección por Constructores contra Code Smells:** Toda la base de código backend migró desde la inyección de atributos oculta `@Autowired` hacia la inyección explícita a través de constructores con variables finales inmutables, facilitando la instanciación de clases limpias sin sobrecargar de complejidad los entornos de control de calidad.

```

@RestController
@RequestMapping("/ventas")
public class VentaController {

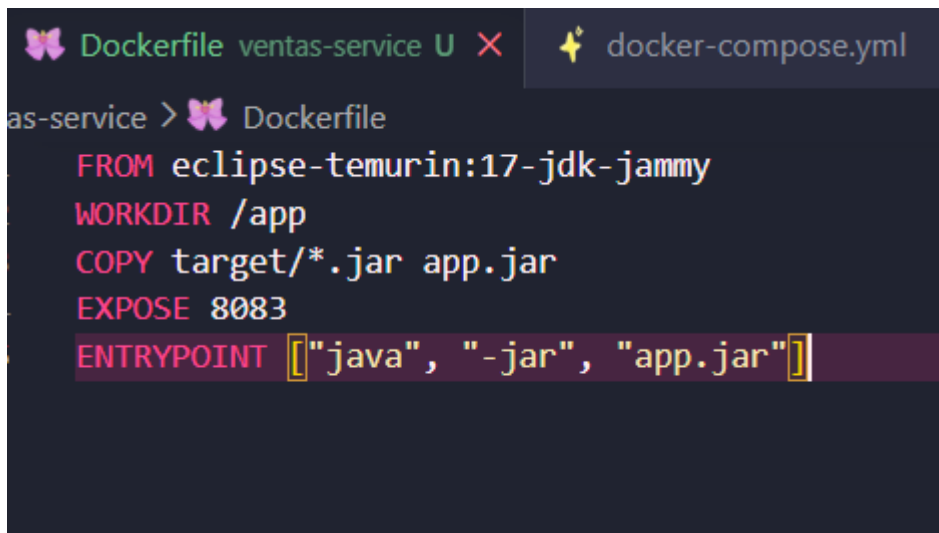
    private final VentaService ventaService;
    public VentaController(VentaService ventaService) {
        this.ventaService = ventaService;
    }
}

```

5. Especificación de Infraestructura y Dockerización (IaC)

Para garantizar un entorno de ejecución portable y consistente, libre de dependencias locales del sistema operativo ("en mi máquina funciona"), se dockerizó el ecosistema completo de Grupo Cordillera. El empaquetamiento se estructuró a través de planos de construcción específicos y una orquestación centralizada.

5.1. Dockerfile para Microservicios Spring Boot



The screenshot shows a code editor with two tabs: 'Dockerfile ventas-service' and 'docker-compose.yml'. The 'Dockerfile' tab is active, showing the following content:

```

FROM eclipse-temurin:17-jdk-jammy
WORKDIR /app
COPY target/*.jar app.jar
EXPOSE 8083
ENTRYPOINT ["java", "-jar", "app.jar"]

```

5.2. Dockerfile para Frontend (React + Nginx Multi-stage)

Para el servidor del Frontend, se implementó una estrategia de **compilación en múltiples etapas (Multi-stage build)** para garantizar que la imagen final de producción solo contenga los archivos estáticos HTML/JS optimizados, reduciendo el tamaño de la imagen en más de un 90%:

```
FROM node:18-alpine as builder

WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .
RUN npm run build
FROM nginx:alpine
COPY --from=builder /app/dist /var/www/grupo-cordillera-frontend
COPY nginx.conf /etc/nginx/conf.d/default.conf
EXPOSE 80 443
CMD ["nginx", "-g", "daemon off;"]
```

5.3. Orquestador Maestro: docker-compose.yml

El aprovisionamiento de las redes virtuales, volúmenes de almacenamiento persistente y las dependencias de encendido de los componentes se unifica en el siguiente manifiesto maestro de Docker Compose:

```
services:

  mysql-db:

    image: mysql:8.0

    container_name: mysql-db

    environment:

      MYSQL_ROOT_PASSWORD: root

      MYSQL_DATABASE: cordillera_db

    ports:
```

```
- "3307:3306"

volumes:

  - ./mysql-data:/var/lib/mysql

healthcheck:

  test: ["CMD", "mysqladmin", "ping", "-h", "localhost"]

  interval: 5s

  timeout: 5s

  retries: 10

networks:

  - cordillera-network


rabbitmq:

  image: rabbitmq:3-management

  container_name: rabbitmq

  ports:

    - "5672:5672"

    - "15672:15672"

  networks:

    - cordillera-network


api-gateway:

  build: ./api-ateway

  container_name: api-gateway

  ports:

    - "8080:8080"
```

```
depends_on:

  mysql-db:

    condition: service_healthy

networks:

  - cordillera-network


usuario-service:

  build: ./usuario-service

  container_name: usuario-service

  ports:

    - "8081:8081"

  environment:

    SPRING_DATASOURCE_URL: jdbc:mysql://mysql-db:3306/cordillera_db

  depends_on:

    mysql-db:

      condition: service_healthy

  networks:

    - cordillera-network


ventas-service:

  build: ./ventas-service

  container_name: ventas-service

  ports:

    - "8083:8083"

  environment:
```

```
    SPRING_DATASOURCE_URL: jdbc:mysql://mysql-db:3306/cordillera_db

depends_on:

  mysql-db:

    condition: service_healthy

  usuario-service:

    condition: service_started

  inventario-service:

    condition: service_started

networks:

  - cordillera-network

inventario-service:

  build: ./inventario-service

  container_name: inventario-service

  ports:

    - "8082:8082"

  environment:

    SPRING_DATASOURCE_URL: jdbc:mysql://mysql-db:3306/cordillera_db

  depends_on:

    mysql-db:

      condition: service_healthy

  networks:

    - cordillera-network

kpi-service:
```

```
build: ./kpi-service

container_name: kpi-service

ports:

  - "8084:8084"

environment:

  SPRING_DATASOURCE_URL: jdbc:mysql://mysql-db:3306/cordillera_db

depends_on:

  mysql-db:

    condition: service_healthy

  ventas-service:

    condition: service_started

  inventario-service:

    condition: service_started

networks:

  - cordillera-network

networks:

  cordillera-network:

    driver: bridge
```

6. Evidencia de Ejecución de Pruebas (BUILD SUCCESS)

A continuación, se adjunta la evidencia de la correcta ejecución de la suite automatizada unificada del proyecto, certificando que los tests unitarios pasaron sin fallos lógicos ni excepciones no controladas:

```
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ usuario-service ---
[INFO] Copying 1 resource from src\main\resources to target\classes
[INFO] Copying 1 resource from src\main\resources to target\classes
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ usuario-service ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ usuario-service ---
[INFO] skip non existing resourceDirectory C:\Users\darly\OneDrive\Escritorio\backend\usuario-service\src\test\resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ usuario-service ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:3.5.2:test (default-test) @ usuario-service ---
[INFO] No tests to run.
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 5.242 s
[INFO] Finished at: 2026-06-18T10:16:59-04:00
[INFO] -----
1. PS C:\Users\darly\OneDrive\Escritorio\backend\usuario-service>
```

2.

```
PS C:\Users\darly\OneDrive\Escritorio\backend\ventas-service> .\mvnw test
[INFO] --- jacoco:0.8.12:prepare-agent (default) @ ventas-service ---
[INFO] argLine set to -javaagent:C:\Users\darly\.m2\repository\org\jacoco\org.jacoco.agent\0.8.12\org.jacoco.agent-0.8.12-runtime.jar=destfile=C:\Users\darly\OneDrive\Escritorio\backend\ventas-service\target\jacoco.exec
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ ventas-service ---
[INFO] Copying 1 resource from src\main\resources to target\classes
[INFO] Copying 0 resource from src\main\resources to target\classes
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ ventas-service ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ ventas-service ---
[INFO] skip non existing resourceDirectory C:\Users\darly\OneDrive\Escritorio\backend\ventas-service\src\test\resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ ventas-service ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ ventas-service ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.cordillera.ventas.VentaControllerTest
WARNING: A Java agent has been loaded dynamically (C:\Users\darly\.m2\repository\net\bytebuddy\byte-buddy-agent\1.14.16\byte-buddy-agent-1.14.16.jar)
WARNING: If a serviceability tool is in use, please run with -XX:+EnabledDynamicAgentLoading to hide this warning
```

```

PS C:\Users\darly\OneDrive\Escritorio\backend\ventas-service> .\mvnw test
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warning
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended
10:18:26.780 [main] INFO org.hibernate.validator.internal.util.Version -- HV000001: Hibernate Validator 8.0.1.Final
10:18:27.980 [main] INFO org.springframework.mock.web.MockServletContext -- Initializing Spring TestDispatcherServlet ''
10:18:27.980 [main] INFO org.springframework.test.web.servlet.TestDispatcherServlet -- Initializing Servlet ''
10:18:27.991 [main] INFO org.springframework.test.web.servlet.TestDispatcherServlet -- Completed initialization in 7 ms
10:18:28.452 [main] INFO org.springframework.mock.web.MockServletContext -- Initializing Spring TestDispatcherServlet ''
10:18:28.453 [main] INFO org.springframework.test.web.servlet.TestDispatcherServlet -- Initializing Servlet ''
10:18:28.453 [main] INFO org.springframework.test.web.servlet.TestDispatcherServlet -- Completed initialization in 0 ms
DEBUG - Valor de metodoPago en solicitud: TARJETA
10:18:28.738 [main] INFO org.springframework.mock.web.MockServletContext -- Initializing Spring TestDispatcherServlet ''
10:18:28.738 [main] INFO org.springframework.test.web.servlet.TestDispatcherServlet -- Initializing Servlet ''
10:18:28.739 [main] INFO org.springframework.test.web.servlet.TestDispatcherServlet -- Completed initialization in 0 ms
10:18:28.768 [main] INFO org.springframework.mock.web.MockServletContext -- Initializing Spring TestDispatcherServlet ''
10:18:28.768 [main] INFO org.springframework.test.web.servlet.TestDispatcherServlet -- Initializing Servlet ''
10:18:28.768 [main] INFO org.springframework.test.web.servlet.TestDispatcherServlet -- Completed initialization in 0 ms
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 5.854 s -- in com.cordillera.ventas.VentaControllerTest
[INFO] Running com.cordillera.ventas.VentaServiceImplTest
[INFO] Tests run: 6, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.937 s -- in com.cordillera.ventas.VentaServiceImplTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 10, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----

```

```

[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 11.705 s
[INFO] Finished at: 2026-06-18T10:18:30-04:00
[INFO] -----
PS C:\Users\darly\OneDrive\Escritorio\backend\ventas-service>

```

3.

```

PS C:\Users\darly\OneDrive\Escritorio\backend\ventas-service> .\mvnw test
[INFO] -----< com.cordillera:inventario-service >-----
[INFO] Building inventario-service 0.0.1-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ inventario-service ---
[INFO] Copying 1 resource from src\main\resources to target\classes
[INFO] Copying 0 resource from src\main\resources to target\classes
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ inventario-service ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ inventario-service ---
[INFO] skip non existing resourceDirectory C:\Users\darly\OneDrive\Escritorio\backend\inventario-service\src\test\resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ inventario-service ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ inventario-service ---
[INFO] No tests to run.
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 3.732 s
[INFO] Finished at: 2026-06-18T10:18:26-04:00
[INFO]
PS C:\Users\darly\OneDrive\Escritorio\backend\inventario-service>

```

4.

```
PS C:\Users\darly\OneDrive\Escritorio\backend\kpi-service> .\mvnw test
rk.security.web.authentication.ui.DefaultLogoutPageGeneratingFilter@76603c44, org.springframework.security.web.authentication.www.BasicAuthentic
ationFilter@135fb22d, org.springframework.security.web.savedrequest.RequestCacheAwareFilter@6a9e88f1, org.springframework.security.web.servletap
i.SecurityContextHolderAwareRequestFilter@591bffd, org.springframework.security.web.authentication.AnonymousAuthenticationFilter@398f6622, org.
springframework.security.web.access.ExceptionTranslationFilter@4daa53e0, org.springframework.security.web.access.intercept.AuthorizationFilter@3
554de46]
2026-06-18T10:18:42.972-04:00 INFO 7964 --- [kpi-service] [          main] o.s.b.t.m.w.SpringBootMockServletContext : Initializing Spring Test
DispatcherServlet ''
2026-06-18T10:18:42.972-04:00 INFO 7964 --- [kpi-service] [          main] o.s.t.web.servlet.TestDispatcherServlet : Initializing Servlet ''
2026-06-18T10:18:42.976-04:00 INFO 7964 --- [kpi-service] [          main] o.s.t.web.servlet.TestDispatcherServlet : Completed initialization
in 3 ms
2026-06-18T10:18:43.058-04:00 INFO 7964 --- [kpi-service] [          main] c.c.k.service.KpiControllerTest : Started KpiControllerTes
t in 6.982 seconds (process running for 9.713)
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 8.764 s -- in com.cordillera.kpiservice.service.KpiControllerTest
[INFO] Running com.cordillera.kpiservice.service.KpiServiceTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.405 s -- in com.cordillera.kpiservice.service.KpiServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 15.418 s
[INFO] Finished at: 2026-06-18T10:18:44-04:00
[INFO] -----
PS C:\Users\darly\OneDrive\Escritorio\backend\kpi-service>
```

5.

```
PS C:\Users\darly\OneDrive\Escritorio\backend\api-atateway> .\mvnw test
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 7.875 s -- in com.cordillera.api_ateway.ApiAtewayApplicationTests
[INFO] Running com.cordillera.api_ateway.AuthenticationFilterTest
GATEWAY DEBUG - Petición recibida en: /private/path
GATEWAY DEBUG - Error: Header Authorization ausente o inválido.
GATEWAY DEBUG - Petición recibida en: /public/path
GATEWAY DEBUG - Ruta pública, pasando...
GATEWAY DEBUG - Petición recibida en: /private/path
GATEWAY DEBUG - Token válido, reenviando a microservicio...
GATEWAY DEBUG - Petición recibida en: /private/path
GATEWAY DEBUG - Error: Token inválido según JwtUtil.
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.829 s -- in com.cordillera.api_ateway.AuthenticationFilterTest
[INFO] Running com.cordillera.api_ateway.JwtFilterTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.008 s -- in com.cordillera.api_ateway.JwtFilterTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 6, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 16.992 s
[INFO] Finished at: 2026-06-18T10:18:48-04:00
[INFO] -----
PS C:\Users\darly\OneDrive\Escritorio\backend\api-atateway>
```

7. Test de Swagger

Responses

Curl

```
curl -X 'GET' \
  'http://localhost:8081/usuarios/1' \
  -H 'accept: */*'
```



Request URL

```
http://localhost:8081/usuarios/1
```

Server response

Code

Details

200

Response body

```
{
  "id": 1,
  "nombreUsuario": "Toshi",
  "email": "toshi@cordillera.com",
  "rol": "ADMIN"
}
```



Download

Response headers

```
cache-control: no-cache,no-store,max-age=0,must-revalidate
connection: keep-alive
content-type: application/json
date: Thu,18 Jun 2026 15:37:02 GMT
expires: 0
keep-alive: timeout=60
pragma: no-cache
transfer-encoding: chunked
vary: Origin,Access-Control-Request-Method,Access-Control-Request-Headers
x-content-type-options: nosniff
x-frame-options: DENY
x-xss-protection: 0
```

Curl

```
curl -X 'GET' \
  'http://localhost:8082/productos/1' \
  -H 'accept: */*'

```



Request URL

```
http://localhost:8082/productos/1
```

Server response

Code

Details

200

Response body

```
{
  "id": 1,
  "nombre": "Gato",
  "precio": 850000,
  "cantidad": 10,
  "description": "PC de alto rendimiento para dise\u00f1o y gaming"
}
```



Download

Response headers

```
cache-control: no-cache,no-store,max-age=0,must-revalidate
connection: keep-alive
content-type: application/json
date: Thu, 18 Jun 2026 15:39:58 GMT
expires: 0
keep-alive: timeout=60
pragma: no-cache
transfer-encoding: chunked
vary: Origin,Access-Control-Request-Method,Access-Control-Request-Headers
x-content-type-options: nosniff
x-frame-options: DENY
x-robots-tag: noindex
```

Responses

Curl

```
curl -X 'PUT' \
  'http://localhost:8082/productos/1/descontar?cantidad=2' \
  -H 'accept: */*'
```



Request URL

```
http://localhost:8082/productos/1/descontar?cantidad=2
```

Server response

Code

Details

200

Response body

```
Stock actualizado correctamente.
```



Download

Response headers

```
cache-control: no-cache,no-store,max-age=0,must-revalidate
connection: keep-alive
content-length: 32
content-type: text/plain;charset=UTF-8
date: Thu,18 Jun 2026 15:40:53 GMT
expires: 0
keep-alive: timeout=60
pragma: no-cache
vary: Origin,Access-Control-Request-Method,Access-Control-Request-Headers
x-content-type-options: nosniff
x-frame-options: DENY
x-xss-protection: 0
```

Responses

Code

Details

201

Undocumented

Response body

```
{
  "id": 4,
  "nombre": "pan",
  "precio": 0.1,
  "cantidad": 10,
  "description": "panconkeso"
}
```



Download

Response headers

```
cache-control: no-cache,no-store,max-age=0,must-revalidate
connection: keep-alive
content-type: application/json
date: Thu,18 Jun 2026 15:41:47 GMT
expires: 0
keep-alive: timeout=60
pragma: no-cache
transfer-encoding: chunked
vary: Origin,Access-Control-Request-Method,Access-Control-Request-Headers
x-content-type-options: nosniff
x-frame-options: DENY
x-xss-protection: 0
```

Responses

Code	Description	Links
200	OK	No links
Media type		
/		

Request URL

`http://localhost:8082/productos`

Server response

Code

Details

200

Response body

```
[{"id": 1, "nombre": "string", "precio": 0.1, "cantidad": 1073741822, "description": "string"}, {"id": 2, "nombre": "Gato", "precio": 850000, "cantidad": 10, "description": "PC de alto rendimiento para dise\u00f1o y gaming"}, {"id": 3, "nombre": "Laptop", "precio": 999.99, "cantidad": 10, "description": "Laptop Gaming"}, {"id": 4, "nombre": "pan", "precio": 0.1, "cantidad": 10, "description": "panconkeso"}]
```



Download

Response headers

```
cache-control: no-cache,no-store,max-age=0,must-revalidate
connection: keep-alive
content-type: application/json
date: Thu,18 Jun 2026 15:42:03 GMT
expires: 0
keep-alive: timeout=60
pragma: no-cache
```

Curl

```
curl -X 'GET' \
  'http://localhost:8082/productos/buscar?nombre=pan' \
  -H 'accept: */*'

```



Request URL

http://localhost:8082/productos/buscar?nombre=pan

Server response

Code

Details

200

Response body

```
[
  {
    "id": 4,
    "nombre": "pan",
    "precio": 0.1,
    "cantidad": 10,
    "description": "panconkeso"
  }
]
```



Download

Response headers

```
cache-control: no-cache,no-store,max-age=0,must-revalidate
connection: keep-alive
content-type: application/json
date: Thu,18 Jun 2026 15:42:34 GMT
expires: 0
keep-alive: timeout=60
pragma: no-cache
transfer-encoding: chunked
vary: Origin,Access-Control-Request-Method,Access-Control-Request-Headers
x-content-type-options: nosniff
x-frame-options: DENY
x-xss-protection: 0

```

Responses

Server response

Code

Details

200

Response body

[Download file](#)

Response headers

```
cache-control: no-cache,no-store,max-age=0,must-revalidate
connection: keep-alive
content-disposition: attachment; filename=reporte_kpi.pdf
content-length: 1303
content-type: application/pdf
date: Thu,18 Jun 2026 15:44:31 GMT
expires: 0
keep-alive: timeout=60
pragma: no-cache
vary: Origin,Access-Control-Request-Method,Access-Control-Request-Headers
x-content-type-options: nosniff
x-frame-options: DENY
x-xss-protection: 0
```

Responses

Code

Description

Links

```
curl -X 'GET' \
  'http://localhost:8084/api/kpi/inventario/estado' \
  -H 'accept: */*'

```

Request URL

http://localhost:8084/api/kpi/inventario/estado

Server response

Code

Details

200

Response body

```
[
  {
    "id": 1,
    "nombre": "string",
    "precio": 0.1,
    "cantidad": 1073741822,
    "descripcion": null
  },
  {
    "id": 2,
    "nombre": "Gato",
    "precio": 850000,
    "cantidad": 10,
    "descripcion": null
  },
  {
    "id": 3,
    "nombre": "Laptop",
    "precio": 999.99,
    "cantidad": 10,
    "descripcion": null
  },
  {
    "id": 4,
    "nombre": "pan",

```